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Western Cape Government
FOR YOU

EDUCATIONAL PROGRAM REPORTING SYSTEM (EPRS)

INF3003W BUSINESS CASE

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Abbreviations Table

Abbreviation	Full Term
API	Application Programming Interface
ASP.NET	Active Server Pages Network Enabled Technologies
Auth	Authorization
Cel	Centre for e-Innovation
C#	C Sharp
CMS	Content Management System
DHET	Department of Higher Education and Training
DSD	Department of Social Development
EPRS	Educational Programme Reporting System
ETL	Extract Transform Load
IT	Information Technology
KPI	Key Performance Indicator
MVP	Minimum Viable Product
PIA	Privacy Impact Assessment
PIR	Project Initiation Request
POPIA	Protection of Personal Information Act
PWA	Progressive Web App
RBAC	Role Based Access Control
SQL	Structured Query Language
UI	User Interface
UML	Unified Modelling Language
UX	User Experience
WCG	Western Cape Government
VPN	Virtual Private Network

Table 1: List of Abbreviations

Introduction

This document presents the business case for the proposed Educational Program Reporting System project undertaken by Team 8 as part of the INF30003W course. The project responds to the problem faced by the Western Cape Government's juvenile facility, where student academic data is currently managed through manual processes that lead to errors, delays, and missed opportunities for performance analysis. This document examines these challenges in detail, outlines the opportunities for improvement, and proposes a feasible solution aligned with the facility's operational needs and strategic objectives. It further assesses the project's feasibility, risks, implementation strategies, quality assurance methods, and alternative solutions to support the justification for developing and deploying the proposed system.

Business Background

The Western Cape Government, through the Western Cape Department of Social Development, is responsible for providing social welfare and support services to vulnerable individuals and communities across the province. One of the department's key responsibilities is the management and support of Child and Youth Care Centres and juvenile facilities that provide care, rehabilitation, protection and developmental services to young offenders, at risk youth, and children awaiting trial. These facilities aim to support education, behavioural development, and reintegration into society.

Currently, many juvenile facilities with the government rely on manual reporting processes to record learner progress, behavioural reports, and educational performance. Information is often stored across multiple systems, spreadsheets or paper-based files making it difficult for staff, members, educators, social workers, and administrators to efficiently access, update and share accurate information. This creates delays in report generation, increases the risk of human error, limits data transparency, and makes tracking even more difficult.

Project Background

The Western Cape Government runs education programmes for young people living in juvenile facilities across the province, through the Centre for e-Innovation (CeI) and the Department of Social Development (DSD) working together. These programmes are an important part of rehabilitation and development, helping learners improve their education and future opportunities. The Educational Programme Reporting System (EPRS) has been proposed to address challenges experienced in the current reporting process.

The Educational Programme Reporting System (EPRS) has been proposed to address challenges experienced in the current reporting process. The DSD in partnership with the CeI, is responsible for delivering and monitoring educational programmes in juvenile facilities across the Western Cape. Their role includes tracking learner progress, ensuring programme effectiveness and producing reports for oversight and planning. Accurate and timely data is critical for these responsibilities, as it supports compliance, performance monitoring, and strategic decision-making.

Existing System

At present, the educational data capturing process is done manually, whereby educators enter marks into Excel spreadsheets and use manual formulas or built-in functions to perform calculations. Afterwards, report cards are compiled by educators who fill in Word document templates. This approach is time-consuming, prone to errors and difficult to standardise across facilities. It also makes aggregation, reporting and compliance with the Protection of Personal Information Act (POPIA) difficult to manage. Additionally, any research attempts including the educational information of juvenile students are difficult to conduct due to the mentioned errors and lack of format standardisation.

The reliance on manual processes limits the Department's ability to accurately track learner progress, generate consolidated reports and use educational data effectively for monitoring and planning purposes. As a result, there is a clear need for a more efficient and reliable solution.

Proposed System

The proposed EPRS is a secure, web-based system that will centralise the capture and management of educational data. It will automate calculations, generate standardised reports and report cards and provide role-based access to sensitive information. The system will also support reporting and analytics to improve oversight and decision-making, while ensuring data accuracy, consistency, and compliance with legislation.

Situations of Business Concern & Opportunities

Business concerns refer to the existing problems, inefficiencies, and risks within the current manual reporting process that negatively impact data accuracy, staff productivity, student tracking, and information security. Opportunities represent the potential improvements and positive outcomes that can be realised by addressing each concern through a well-designed digital reporting system. The table below outlines each identified concern, describes the opportunity that arises from resolving it, and highlights the benefit to the department from addressing each concern.

Situations of concern	Benefits from addressing concern	Opportunity arising from addressing concern
Inefficient and error-prone manual data capturing	• Improved accuracy of marks and records • Faster data entry and report generation • Reduced workload for teachers • Fewer duplicated or inconsistent records • Easier correction of mistakes • More efficient preparation of student reports and mark schedules.	• Introduction of a centralised digital mark capturing process • Automated calculation of assessment results using configured weightings • Excel import and export functionality to simplify bulk mark capturing • Implementation of validation rules to prevent invalid or incomplete entries • Automated generation of academic reports and mark schedules
Lack of system integration across departments	• Centralised data access • Consistent information across departments • Reduced duplication of work • Improved communication between staff	• Development of a unified educational reporting platform • Role-based access to relevant academic records and management information • Standardisation of assessment configuration and reporting processes

	• Streamlined administrative processes	• Improved coordination between facility-level and departmental users • Establishment of a common database for academic records
Difficulty tracking performance overtime	• Easy access to historical student data • Clear tracking of progress and trends • Early identification of struggling students • Better academic planning • More accurate reporting	• Development of individual student profiles containing academic information • Implementation of the Predictive Watchlist to identify students who may require additional academic support • Access to archived academic reports and mark schedules • Development of academic progression functionality • Improved monitoring of academic performance through reporting and analytics
Lack of data analysis	• Access to performance statistics (averages, pass rates) • Better understanding of subject performance • Improved decision-making • Identification of gaps in learning • More structured reporting	• Introduction of dashboards and analytical reporting functionality • Generation of performance summaries and management reports • Analysis of assessment results and academic performance • Use of visual charts to communicate performance information • Greater use of academic data to inform educational planning and support interventions
Security and Privacy Risk(Compliance Challenges)	• Secure storage of student data	• Implementation of role-based access controls and permission management

	• Controlled access to sensitive information • Compliance with data protection regulations • Reduced risk of data breaches	• Introduction of audit logging to record relevant system activities • Restriction of sensitive functionality to authorised users • Centralised management of staff accounts and access permissions • Establishment of a foundation for improved information governance and future security enhancements

Table 2: Business Opportunities and Concerns

System Objectives

System objectives define what the proposed educational reporting system must achieve to address the concerns identified. They translate the project's purpose into specific, measurable, and actionable statements that guide development, testing, and final evaluation. Each objective is written as a clear outcome that can be verified upon completion. The following objectives have been established for this project.

- Digitise all student academic data entry processes through structured online forms, including student registration, assessment configuration, and mark capturing, with built-in validation rules to reduce data capturing errors by at least 80% and eliminate reliance on paper-based and spreadsheet-only reporting processes by the end of June.
- Configure the system to automatically calculate termly and yearly academic results using predefined assessment weightings, reducing manual calculation errors by at least 95% and saving educators a minimum of 60% of the time previously spent capturing marks and preparing academic reports by mid-July.
- Ensure the system can generate individual and batch academic reports for all registered students with available assessment records, including assessment results and educator comments, reducing report preparation time by at least 70% and allowing authorised users to generate and export reports in PDF format within 2 minutes by the end of June.
- Provide centralised access to student academic records, including historical assessment results, archived reports, and academic progression information, enabling authorised users to retrieve student information within 30 seconds and monitor academic performance across multiple terms and years upon deployment.
- Implement a Predictive Watchlist capable of identifying students at risk of poor academic performance based on their assessment results, with the aim of identifying at least 80% of students who meet the defined academic risk criteria and supporting earlier intervention by educators and management by the end of July.
- Embed analytical dashboards and management reporting tools that provide performance summaries, subject-level results, and academic trends, reducing the time required to compile academic performance statistics by at least 60% and enabling authorised management users to access updated performance information within 1 minute by early August.
- Improve the security of student academic information by implementing role-based access controls, permission management, and audit logging, ensuring that 100% of protected system functions are restricted to authorised users and that designated changes to sensitive academic records are recorded, with zero critical

access-control vulnerabilities identified during final security testing by September.

- Provide integrated user support through the Teachers Pet help assistant, system guidance, and IT support request functionality, with the aim of enabling users to resolve at least 70% of common system navigation difficulties without direct assistance from the development team and reducing the time required to report technical issues by at least 50% by September.